

Math Camp 2026 — Day 3 Exercises: Solutions

Zhilin Lu

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Part I — Probability

1. Reading a table

The table has 300 countries. Democracies: 120. Compliers: 168. Both: 96. Filling in the rest of the 2×2 is worth doing at the board before answering anything, because most of the confusion in this question dissolves once students can see all four cells:

	Complied	Did not	Total
Democracy	96	24	120
Non-democracy	72	108	180
Total	168	132	300

(a) Every one of these is a cell (or a margin) divided by the grand total:

$$P(\text{Dem} \cap \text{Comp}) = \frac{96}{300} = 0.320, \quad P(\text{Dem}) = \frac{120}{300} = 0.400, \quad P(\text{Comp}) = \frac{168}{300} = 0.560$$

The joint probability asks: pick a country at random from all 300 — what is the chance it is both? The marginals ask the same question about one attribute, ignoring the other. All three share the same denominator, 300, which is exactly what makes them *unconditional*.

(b) Conditioning changes the denominator. That sentence is the whole lesson of the question.

$$P(\text{Comp} \mid \text{Dem}) = \frac{96}{120} = 0.800, \quad P(\text{Dem} \mid \text{Comp}) = \frac{96}{168} = 0.571$$

The first restricts attention to the top row — we have been *told* the country is a democracy, so the 180 non-democracies are no longer in the sample space — and asks what share of that row complied. The second restricts attention to the “Complied” column and asks what share of it was democratic. Same numerator, different worlds.

Formally this is just the definition:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.320}{0.400} = 0.800$$

Watch for: students reporting the second part of (b) as 0.800 — conditioning in the direction the sentence *sounds* like rather than the direction it says. This is the single most common error on the page, and it is not a computational slip; it is the error that Bayes’ rule in Question 2 exists to correct. Flag it here so that Question 2 lands as the fix for a mistake they have already made rather than as a new formula.

A useful diagnostic to say out loud: “Of democracies, how many complied?” versus “Of compliers, how many were democracies?” If a student can restate the question in that “of , *how many* ” form, they will pick the right denominator every time. The phrase after “of” is the denominator.

(c) Independence would mean that knowing the regime type tells you nothing about compliance — that the joint factors into the product of the marginals:

$$P(\text{Dem})P(\text{Comp}) = (0.400)(0.560) = 0.224 \neq 0.320$$

So the two are **not** independent. The joint probability exceeds what independence predicts (0.320 vs. 0.224, a ratio of about 1.43), which means regime type and compliance are *positively* associated.

The conditional probabilities show the same thing more vividly: 80% of democracies complied versus $72/180 = 40\%$ of non-democracies. An equivalent test of independence is whether $P(\text{Comp} | \text{Dem}) = P(\text{Comp})$ — here, $0.800 \neq 0.560$, so no. Any of the three tests works; they are algebraically the same statement.

Worth saying explicitly, because it is where this exercise connects to the rest of their training: association is not causation. Nothing here rules out that democracies are also richer, more trade-dependent, or more closely allied to the sanctioning state, and that one of those is doing the work.

2. Bayes

Setup: 2% of containers are illicit. The scanner flags 90% of illicit containers (sensitivity) and 5% of legitimate ones (false positive rate).

(a) The denominator is the total probability of a flag, which arrives through two disjoint channels — a true flag and a false one:

$$P(F) = \underbrace{(0.90)(0.02)}_{\text{illicit and flagged}} + \underbrace{(0.05)(0.98)}_{\text{legit and flagged}} = 0.018 + 0.049 = 0.067$$

$$P(I | F) = \frac{P(F | I)P(I)}{P(F)} = \frac{0.018}{0.067} = 0.269$$

So a flagged container is illicit about 27% of the time — meaning roughly three out of four flags are false alarms, from a test that is 90% sensitive and 95% specific.

It is worth walking this through a second time in natural frequencies, which nearly everyone finds easier than the formula. Imagine 1,000 containers:

- 20 are illicit; the scanner catches 18 of them.
- 980 are legitimate; the scanner falsely flags 49 of them.

- Total flags: $18 + 49 = 67$. True ones: 18. So $18/67 = 27\%$.

Same arithmetic, no notation. If a student is stuck on the formula, have them draw the tree with counts.

(b) The intuition: illicit containers are rare, so the 5% false-positive rate is being applied to a *very large* group. Ninety-eight percent of containers are legitimate, and 5% of a big number beats 90% of a small one — 49 false flags against 18 true ones.

The general point is that the number of false positives is driven by the size of the group being falsely flagged, not by the accuracy of the test. Improving sensitivity from 90% to 99% would barely move the answer (it would raise true flags from 18 to about 20). Cutting the false-positive rate from 5% to 1% would transform it: 18 true flags against 10 false ones, i.e. $P(I | F) = 0.65$. When the base rate is low, specificity is the binding constraint, and everyone's instinct is to optimize the other one.

This is the same structure as the medical-screening example students will have seen, and the same structure as any “we flagged 10,000 suspicious transactions” claim they will read in a policy report. The question to ask such a claim is always: flagged out of how many, and what is the base rate?

(c) Now the prior is 20% rather than 2% — a targeted screen of containers from a suspect route, say, rather than everything at the port.

$$P(F) = (0.90)(0.20) + (0.05)(0.80) = 0.180 + 0.040 = 0.220, \quad P(I | F) = \frac{0.180}{0.220} = 0.818$$

The test did not change. Nothing about the machine is different. Only the population it was pointed at changed, and the posterior went from 27% to 82%.

The lesson: a tool that is nearly useless applied broadly can be highly informative applied to a population already enriched for what you are looking for. That is the general argument for targeted screening — and, incidentally, the general argument for having a theory before you go looking for evidence.

Optional, if the group is comfortable: the odds form makes the structure much clearer and is worth ten minutes if you have them.

$$\underbrace{\frac{P(I | F)}{P(\neg I | F)}}_{\text{posterior odds}} = \underbrace{\frac{P(I)}{P(\neg I)}}_{\text{prior odds}} \times \underbrace{\frac{P(F | I)}{P(F | \neg I)}}_{\text{likelihood ratio}}$$

The likelihood ratio is a property of the test alone: $0.90/0.05 = 18$. The test multiplies your prior odds by 18, always, regardless of the base rate.

- Part (a): prior odds $2/98 = 0.0204$; posterior odds $0.0204 \times 18 = 0.367$; probability $0.367/1.367 = 0.269$. ✓
- Part (c): prior odds $20/80 = 0.25$; posterior odds $0.25 \times 18 = 4.5$; probability $4.5/5.5 = 0.818$. ✓

Seeing that the *same* multiplier produces both answers is the cleanest way to make the point that (b) and (c) are about the prior, not the test.

3. Expectation and variance

The distribution is $X \in \{1, 2, 3, 4\}$ with probabilities 0.3, 0.4, 0.2, 0.1.

(a) Expectation is the probability-weighted sum of the values:

$$E[X] = 1(0.3) + 2(0.4) + 3(0.2) + 4(0.1) = 0.3 + 0.8 + 0.6 + 0.4 = 2.100$$

For $E[X^2]$, weight the *squared* values by the same probabilities. Note carefully that the probabilities do not get squared — only the values do:

$$E[X^2] = 1(0.3) + 4(0.4) + 9(0.2) + 16(0.1) = 0.3 + 1.6 + 1.8 + 1.6 = 5.300$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 5.300 - 4.410 = 0.890, \quad SD(X) = \sqrt{0.890} = 0.943$$

A sanity check students should internalize: the SD of 0.943 is a plausible “typical distance from 2.1” for a variable that lives on $\{1, 2, 3, 4\}$. If they compute an SD of 5 for this distribution, something has gone wrong, and they should be able to see that without checking the algebra. Variance carries squared units, which is why it is hard to eyeball; the SD is back in the units of X , which is why we report it.

(b) The gap is $5.300 - 4.410 = 0.890$, which is precisely the variance — that is exactly what $\text{Var}(X) = E[X^2] - (E[X])^2$ says, rearranged.

The deeper point is that $E[X^2] \neq (E[X])^2$: the average of the squares is not the square of the average. This is an instance of the general fact that $E[g(X)] \neq g(E[X])$ for non-linear g (Jensen’s inequality; for convex g like squaring, $E[g(X)] \geq g(E[X])$, with equality only if X is constant). Squaring is convex, so the average of the squares is pulled up by the large values more than it is pulled down by the small ones, and the size of that gap *is* the spread.

This is not a technicality they can file away. It is why the mean of a ratio is not the ratio of the means, why average per-capita growth across countries is not growth of the average, and why you cannot back out $E[\log Y]$ from $\log E[Y]$. Expectations pass through linear functions and nothing else.

(c) For $Y = 5X - 3$:

$$E[Y] = 5E[X] - 3 = 5(2.100) - 3 = 7.500$$

$$\text{Var}(Y) = 5^2 \text{Var}(X) = 25(0.890) = 22.250, \quad SD(Y) = \sqrt{22.250} = 4.717$$

Two things to draw out. First, expectation is fully linear — both the multiplier and the constant

pass through. Second, variance is not: the -3 shifts the whole distribution left and contributes nothing to the spread, since shifting every value by the same amount leaves all distances between values unchanged. Only the multiplicative constant matters, and it enters squared.

Note that $SD(Y) = 5 \times SD(X) = 5(0.943) = 4.717$, which is the tidier way to see it: the SD scales by $|a|$, the variance by a^2 . The absolute value matters — multiplying by -5 would give the same variance, since spread has no direction.

4. Variance of sums, and correlation

Given $\text{Var}(X) = 4$, $\text{Var}(Y) = 9$, $\text{Cov}(X, Y) = -2$.

(a)

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y) = 4 + 9 + 2(-2) = 9$$

$$\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2 \text{Cov}(X, Y) = 4 + 9 - 2(-2) = 17$$

$$\rho = \frac{\text{Cov}(X, Y)}{SD(X) \cdot SD(Y)} = \frac{-2}{2 \times 3} = -0.333$$

The correlation is the covariance stripped of units. Covariance is in units of X times units of Y — meaningless on its own, and it changes if you switch GDP from dollars to billions. Dividing by both SDs forces it into $[-1, 1]$ and makes it comparable across variables. That is the only reason ρ exists.

Since the covariance here is negative, the two variables move in opposite directions on average: adding them lets the movements partly cancel (variance falls to 9, less than $\text{Var}(Y)$ alone), while differencing them makes the movements reinforce (variance rises to 17).

(b) Only $E[X + Y] = E[X] + E[Y]$ survives without the covariance term. None of the variance results do.

Expectation ignores dependence entirely — it is linear, full stop, and $E[X + Y] = E[X] + E[Y]$ holds whether X and Y are independent, identical, or perfectly opposed. Variance does not have this luxury, because it is built from squares, and the cross term $2 \text{Cov}(X, Y)$ does not vanish unless the covariance is zero.

Also worth flagging: $\text{Var}(X - Y) = 17$ exceeds either variance alone. The minus sign attaches to the covariance term, not to the variances — variances *add* even under subtraction. This is why estimating a difference is noisier than estimating either mean, and why a difference-in-means with two noisy groups can have a standard error larger than either group's. Students meet this again the moment they compute a treatment effect.

(c) Here X is uniform on $\{-2, -1, 0, 1, 2\}$ and $Y = X^2$.

$$E[X] = 0 \text{ (by symmetry),} \quad E[Y] = \frac{4 + 1 + 0 + 1 + 4}{5} = 2$$

$$E[XY] = E[X \cdot X^2] = E[X^3] = \frac{-8 + (-1) + 0 + 1 + 8}{5} = 0$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - (0)(2) = 0$$

So the covariance, and therefore the correlation, is exactly zero. But Y is a *deterministic function* of X — knowing X tells you Y with certainty. They could hardly be more dependent.

The formal check: $P(Y = 4) = 2/5 = 0.4$, while $P(Y = 4 \mid X = 0) = 0$. Independence would require these to be equal, since independence means conditioning on X changes nothing about Y 's distribution. It plainly changes everything.

The takeaway: zero correlation rules out a *linear* relationship and nothing else. Correlation is a measure of linear association only, and it is blind to every U-shape, threshold, and interaction. Independence implies zero covariance; zero covariance does not imply independence. The arrow runs one way.

Draw the five points on the board — they form a perfect parabola — and ask what a regression line through them would look like. Flat. This is the strongest argument available for plotting your data before modelling it, and it costs thirty seconds to make. (It is also the honest answer to “why did the correlation matrix show nothing?”)

5. Distributions and the CLT

(a) Binomial with $n = 8$, $p = 0.25$:

$$P(X = 2) = \binom{8}{2} (0.25)^2 (0.75)^6 = 28 \times 0.0625 \times 0.1780 = 0.311$$

The three pieces are worth naming separately: $\binom{8}{2} = 28$ counts the ways to choose *which* two of the eight trials succeed, $(0.25)^2$ is the probability that those two succeed, and $(0.75)^6$ is the probability that the other six fail. Multiply because trials are independent; multiply by 28 because we do not care which two.

For the second part, go through the complement rather than summing eight terms:

$$P(X \geq 1) = 1 - P(X = 0) = 1 - (0.75)^8 = 1 - 0.100 = 0.900$$

“At least one” almost always means “one minus none.” It is the single most useful trick in this section.

```
dbinom(2, size = 8, prob = 0.25)      # 0.3115
1 - dbinom(0, size = 8, prob = 0.25)  # 0.8999
```

(b) The Poisson distribution assumes the mean equals the variance — a single parameter λ governs both. Here the variance is nearly four times the mean, so that assumption is badly violated. This is **overdispersion**, and it is the usual case with real count data rather than the exception.

The consequence is not merely aesthetic. If you fit a Poisson and the true variance is four times what the model assumes, the model thinks the data are far more informative than they are: standard errors come out far too small, t statistics too large, confidence intervals too narrow, and significance claims far too generous. You will find effects that are not there.

Why does real count data overdispense? Usually unobserved heterogeneity — the units differ in ways you have not measured, so the rate varies across them rather than being common — or contagion, where one event makes the next more likely. Both are the norm in political data: protests cluster, conflicts recur, some countries are simply more sanction-prone than the covariates reveal.

Fixes, roughly in order of preference: a **negative binomial** model (which adds a dispersion parameter and is the usual first move), **quasi-Poisson** (which keeps the Poisson mean structure but inflates the standard errors by an estimated factor), or robust/cluster-robust standard errors. Check for the problem before choosing: compare the residual deviance to the residual degrees of freedom, and be suspicious if the ratio is much above 1.

(c) Normal with $\mu = 48$, $\sigma = 6$.

$$z = \frac{60 - 48}{6} = 2 \implies P(X > 60) = 1 - 0.977 = 0.023$$

The z score says 60 is two standard deviations above the mean; the tail beyond 2σ holds about 2.3% of the mass, which is half of the familiar 5% outside $\pm 2\sigma$.

For the 95th percentile, run it backwards — start from the probability and solve for the value:

$$x = \mu + z_{0.95}\sigma = 48 + 1.645(6) = 57.87$$

```
1 - pnorm(60, mean = 48, sd = 6) # 0.02275
qnorm(0.95, mean = 48, sd = 6)   # 57.869
```

The p/q distinction is worth stating once and referring back to all week: **p** functions go from a value to a probability, **q** functions go from a probability back to a value. They are inverses. Nearly every early error with these functions is calling one when you wanted the other, and the giveaway is that the answer comes back on the wrong scale — a probability where you expected a test score.

Note also 1.645 versus 1.96: 1.645 leaves 5% in *one* tail, 1.96 leaves 5% split across *two*. Students who have memorized 1.96 will reach for it here.

(d)

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{36}} = \frac{12}{6} = 2$$

To halve the standard error you need $\sqrt{n}$ to double, so n must **quadruple**: $n = 144$.

This $\sqrt{n}$ relationship is the most practically consequential fact in the whole day. Precision is expensive and gets more expensive: going from 100 to 400 respondents buys you the same proportional gain as going from 400 to 1,600. It explains why national surveys plateau around 1,000–1,500 re-

spondents, why the marginal value of another wave of data collection falls so fast, and why a study with 30 observations cannot be rescued by finding 10 more.

(e)

i. False. The CLT is about the sampling distribution of a *statistic* — the mean, across hypothetical repeated samples — not about the distribution of X itself. The underlying variable can be arbitrarily skewed, bimodal, or discrete, and it stays that way no matter how large n gets. Collecting more data does not make your variable normal; it makes your *estimate of its mean* normally distributed. Students conflate these constantly, and the tell is when they check a histogram of the raw variable for normality before running a regression.

ii. True. That is the second of the CLT’s three claims: $\bar{X}$ is centred on μ . (The three together: the sampling distribution is approximately normal, centred on μ , with standard deviation $\sigma/\sqrt{n}$.) Centring on the truth is unbiasedness, and it is a separate property from the normal shape — it holds at any n , whereas normality is asymptotic.

iii. False. The CLT requires independent draws, and students within a classroom are not independent — they share a teacher, a curriculum, a neighbourhood, and a peer group. Their outcomes are positively correlated, so each additional student from the same classroom carries much less new information than a student from a fresh classroom would.

The effective sample size here is closer to 3 (the number of classrooms) than to 36 (the number of students), and the usual formula, which divides by $\sqrt{36}$, will produce standard errors that are far too small. The remedy is to cluster the standard errors at the classroom level, or to treat the classroom as the unit of analysis.

The general rule, and it is worth writing on the board: the n that matters is the number of *independent* units, not the number of rows in your data frame. Anyone working with country-year panels — which is to say, most of the room — should hear this now rather than in their second-year paper.

Part II — Computing

6. Inside or outside `aes()`

(a) Three lines, three different fates, and the middle one is the dangerous case.

Line 1 errors immediately: `Invalid colour: region`. Outside `aes()`, ggplot expects a literal colour — “red”, “steelblue”, a hex code — and it goes looking for a colour named “region”, which does not exist. The error is loud and immediate, which makes this the *good* failure.

Line 2 fails silently. A plot appears, the points are red rather than steelblue, and there is a one-entry legend labelled “steelblue”. Inside `aes()`, ggplot treats the string as a variable with a single category, assigns that category the first default colour (red), and helpfully documents it with a legend. No error, no warning, wrong figure. If you only glance at the output, you may not notice for a long time.

Line 3 is correct: `color = region` inside `aes()` gives four regions, four colours, and an informative legend.

The pattern to name explicitly: the loud failure is the harmless one and the silent failure is the costly one. That asymmetry recurs throughout data work — the errors that stop you are far kinder than the ones that let you keep going. Tell students to be *more* suspicious when code runs clean than when it throws, and to treat a legend they did not ask for as a symptom.

(b) The rule, and it is the only thing from this question they need to retain:

If the property **varies with the data**, it goes **inside** `aes()`. If it is a **constant**, it goes **outside**, as an argument to the geom.

`aes()` stands for aesthetic *mapping* — a mapping from a variable to a visual property. If there is no variable, there is no mapping, and it does not belong there.

The same rule governs `size`, `shape`, `alpha`, `fill`, and `linewidth`. `size = 3` means every point is size 3; `size = population` means point size encodes population.

(c)

```
ggplot(analysis, aes(gdp_pc, vote_share, color = region)) +  
  geom_point()
```

```
ggplot(analysis, aes(gdp_pc, vote_share)) +  
  geom_point(color = "steelblue")
```

7. Lines and panels

(a) You get one jagged line zigzagging across the plot. `geom_line()` connects observations in order of x , and nothing in the code told it which rows belong to the same country, so all 312 rows were joined into a single path. At the end of each country's series it jumps back to 1995 to start the next one, which is what produces the sawtooth.

The underlying issue is that the data are grouped — 12 countries $\times$ 26 years — but the *plot* has no idea, because grouping in the data frame is not grouping in the aesthetic. Any time a line plot comes out looking like a seismograph, this is why.

(b)

```
ggplot(analysis, aes(year, gdp_pc, group = country)) +  
  geom_line(alpha = 0.4)
```

`group = country` tells ggplot to start a fresh line for each country. Note that `group` is the minimal fix: it separates the lines without spending a visual channel on it. Mapping `color = country` would also work and would implicitly group, but it burns the colour channel on 12 categories — more than the eye can distinguish — and produces an unreadable legend. Use `group` when you need separation, `color` when you need identification.

The `alpha = 0.4` is doing real work here: with a dozen overlapping series, partial transparency lets the reader see where lines pile up. It sits outside `aes()` because it is a constant, which is a nice callback to Question 6.

(c) Add `+ facet_wrap(~ region)` to split the plot into one panel per region.

The `scales = "free_y"` question is the substantive one. With free scales you gain the *shape* of the trend within each panel — a region with small absolute values is no longer flattened into a horizontal line at the bottom of a shared axis. You lose the ability to compare *levels* across panels: two panels can look nearly identical while differing by an order of magnitude, because each has been silently rescaled to fill its own frame.

Neither choice is wrong. Free scales are legitimate when the question is about within-group dynamics; fixed scales are right when the question is about differences in level. Free scales *without saying so in the caption* are not legitimate, because the reader’s default assumption is a common axis and the figure gives them no cue otherwise.

Ask the group which they would choose for “did growth accelerate after 2008 in each region?” (free — you want the shape) versus “which region is richest?” (fixed — you want the level). The choice follows from the question, not from which looks better.

8. A pipeline into a plot

(a)

```
analysis |>
  group_by(region, year) |>
  summarize(mean_gdp_pc = mean(gdp_pc, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(x = year, y = mean_gdp_pc, color = region)) +
  geom_line(linewidth = 0.9)
```

Two details worth pausing on. Grouping by `region` *and* `year` collapses the country-level data to one row per region-year, which is exactly what a line per region requires. And `.groups = "drop"` removes the leftover grouping — without it, `summarize()` leaves the data grouped by `region`, and the next operation silently inherits that grouping. That silent inheritance is a classic source of bugs a few steps later, and the message R prints about it is easy to train yourself to ignore.

(b) You pipe *up to* `ggplot()` and then switch to `+` for everything after it. The reason is a historical accident, not a principle: `ggplot2` predates the pipe by years and had already claimed `+` to mean “add another layer.” There is no deeper logic, and it is worth saying so plainly rather than letting students hunt for one. Mixing them up — using `|>` between geoms — produces an error that is not especially informative, so it is better to know the rule than to derive it.

(c) `Viridis` is designed to remain distinguishable under the common forms of colour vision deficiency, which affects roughly 8% of men — meaning that in a room of 30 seminar participants, someone probably cannot read a red-green figure. It also varies **monotonically in lightness**, so the ordering survives greyscale printing and photocopying, and so the perceptual distance between colours matches the numeric distance between values. The default rainbow-style palettes fail on all three counts: they have bright bands in the middle that read as discontinuities, they collapse under greyscale, and they are actively misleading about magnitude.

9. Labelling and saving

(a)

```
p <- ggplot(analysis, aes(gdp_pc, vote_share)) +
  geom_point(alpha = 0.4) +
  scale_x_log10() +
  labs(
    title = "Income and incumbent support",
    subtitle = "12 countries, 1995-2020",
    x = "GDP per capita, USD (log scale)",
    y = "Incumbent vote share (%)",
    caption = "Source: simulated data for teaching"
  ) +
  theme_minimal(base_size = 12)
```

Assigning to `p` rather than printing directly is the habit to build: it lets you save the object, modify it later by adding layers, and reuse it. The axis labels carry units and note the log scale, because a reader who does not know the axis is logged will misread every distance on it. The caption states the source. None of this is decoration — a figure that goes into a paper is read without you standing next to it.

(b)

```
ggsave("output/fig1.pdf", p, width = 6.5, height = 4.5)
ggsave("output/fig1.png", p, width = 6.5, height = 4.5, dpi = 300)
```

PDF is vector — infinitely scalable, no resolution to specify, and the right choice for anything going into LaTeX or a print journal. PNG is raster, which is why it needs `dpi = 300`; use it for slides, the web, and anywhere a vector format is not accepted. The `dpi` argument does nothing for the PDF.

(c) This is the counterintuitive one, and students routinely do the opposite of the fix.

Text size in ggplot is **absolute, not relative**: `base_size = 12` means 12 points, in both files. What changes between a 4-inch canvas and a 10-inch canvas is not the text but the panel around it. On the small canvas, 12pt text is large relative to the panel; on the large canvas, that same 12pt text occupies a much smaller fraction of the figure. Then the journal or slide scales the whole figure down to fit a column, shrinking the text further — and by the time it is printed it is illegible.

So the fix for labels that came out too small is a **smaller canvas**, not a bigger one — or, equivalently, a larger `base_size`. The instinct to “make the figure bigger so the text is bigger” is exactly backwards, because the text was never the thing that changed.

The rule of thumb: save figures at roughly the size they will be displayed. If a figure will occupy a 3-inch column, export it 3 inches wide, and it will need no scaling at all.

10. A Quarto document that will not render

(a) Both options run the chunk; they differ in what the reader sees.

`include: false` runs the chunk and shows **nothing** — no code, no output. This is for the setup chunk: loading libraries, reading data, setting options. The work needs to happen, but `library(tidyverse)` and its startup messages are not part of what you are communicating.

`echo: false` runs the chunk and hides **only the code**, showing the output. This is for figures and tables in a problem set or a paper: the reader should see the plot, not the twelve lines that produced it.

The pair to remember alongside them is `eval: false`, which shows the code without running it — useful when demonstrating syntax you do not want executed. Between the three you can independently control running, showing code, and showing output, which is most of what chunk options are for.

(b) Rendering starts a **brand new R session**. Nothing you created by hand in the console exists there — not the objects in your environment, not the packages you loaded interactively, not the working directory you set. Quarto only knows what is written in the document.

So the `readRDS()` line that lives only in your console needs to move into the setup chunk. This is the single most common reason a document that “works” fails to render, and the fix is nearly always the same: something exists in your session that was never written down.

The diagnostic habit worth teaching: restart R and run the whole document from a clean session, regularly, rather than at the end. The longer you go without doing it, the more invisible dependencies accumulate, and the harder it is to find which one broke it.

(c) If it renders, someone else can run it and get exactly your output. That is what reproducible means — not a philosophy, a property you can check.

The value of Quarto here is that it makes the check automatic and constant. A document that renders from a clean session has *proved* that every step from raw data to final figure is written down somewhere. You cannot have silently depended on an object you made by hand in March and forgot about, because the fresh session would not have it. The alternative — copying numbers into a Word file and hoping you can reconstruct where they came from — fails precisely when it matters most, which is when a reviewer asks eight months later.

Worth adding out loud, because most of them are about to start a project: the person this protects most often is you, six months from now, and you will have forgotten more than you expect.